오후 7:59 11월 12일 토요일 33%
컴퓨터 비전
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Problem set - 02

1. Prove that both the 2-D continuous and discrete Fourier transforms are linear operations (see Section 2.6.2 for a definition of linearity). (10 points)

Let $\mathcal{F}$ is transfer function
and let space variable t, z
and frequency variable u, v
according to linearility of fourier transform

$$a_1 x_1(t) + a_2 x_2(t) \overset{\mathcal{F}}{\Longleftrightarrow} a_1 X_1(f) + b X_2(f)$$

$$\mathcal{F}[a_1 f_1(t,z) + a_2 f_2(t,z)]$$
$$= a_1 \mathcal{F}[f_1(t,z)] + a_2 \mathcal{F}[f_2(t,z)]$$
$$F(u,v) = \mathcal{F}(f(t,z)) = \iint_{-\infty}^{\infty} f(t,z)\, e^{-j2\pi(ut+vz)}\, dt\, dz$$

Then,

$$\mathcal{F}[a_1 f_1(t,z) + a_2 f_2(t,z)] = \iint_{-\infty}^{\infty} [a_1 f_1(t,z) + a_2 f_2(t,z)]\, e^{-j2\pi(ut+vz)}\, dt\, dz$$

$$= a_1 \iint_{-\infty}^{\infty} f_1(t,z) \cdot e^{-j2\pi(ut+vz)}\, dt\, dz + a_2 \iint_{-\infty}^{\infty} f_2(t,z)\, e^{-j2\pi(ut+vz)}\, dt\, dz$$

$$\mathcal{F}[a_1 f_1(t,z) + a_2 f_2(t,z)] = \underline{a_1 \mathcal{F}[f_1(t,z)] + a_2 \mathcal{F}[f_2(t,z)]}$$

so, we can find It is linear operation in continuous transform

In 2D discrete Transform

$$\mathcal{F}(f(t,z)) = \sum_{t=0}^{M-1} \sum_{z=0}^{N-1} f(t,z)\, e^{-j2\pi(ut/M + vz/N)}$$

$$\mathcal{F}[a_1 f_1(t,z) + a_2 f_2(t,z)] = \sum_{t=0}^{M-1} \sum_{z=0}^{N-1} [a_1 f_1(t,z) + a_2 f_2(t,z)]\, e^{-j2\pi[ut/M + vz/N]}$$

$$\Rightarrow \underline{a_1 \mathcal{F}[f_1(t,z)] + a_2 \mathcal{F}[f_2(t,z)]}$$

so, it is same form with continuous transform, we can find

Both continuous and discrete fourier transform satisfy linear operation

3. Show that the saturation component of the complement of a color image cannot be computed from the saturation component of the input image alone. (10 points) ppt. 6 , CH. 6

saturation component of HSI image in terms to RGB component following the equation

$$H = \cos^{-1}\left[\frac{\frac{1}{2}[(R-G)+(R-B)]}{\sqrt{(R-G)^2+(R-B)(G-B)}}\right], \quad S = 1 - \frac{3}{R+G+B}[\min(R,G,B)], \quad I = \frac{1}{3}(R+G+B)$$

we can see S can be inferenced many different RGB color are possible for which the saturation value remain same and increase

for example, suppose the two RGB color (0, 0, 1) , (0, 0, 0.1) , and the corresponding HSI values are (0.67, 1.0, 0.33) , (0.67, 1.0, 0.033)

The complement of RGB colors are (1, 1, 0) , (1, 1, 0.9) , corresponding HSI value will be (0.17, 1.0, 0.33) , (0.17, 0.1, 0.33) .

we can see both saturation of color is 1, complement saturation is different (1 , 0.1) Thus, two different shades of same color can having different complement saturation.

4. The arithmetic decoding process is the reverse of the encoding procedure. Decode the message 0.23355 given the following coding model. (10 points)

Ch. 8

Symbol	Probability	Range
a	0.2	[0, 0.2)
e	0.3	[0.2, 0.5)
i	0.1	[0.5, 0.6)
o	0.2	[0.6, 0.8)
u	0.1	[0.8, 0.9)
!	0.1	[0.9, 1.0]

key point : 1. entropy encoding
2. Shannon's source coding theorem

decoding the message is 0.23355, we can guess first word is e

[0,1]
e [0.2, 0.5)
a [0.2, 0.26]
i [0.23, 0.236]
i [0.233, 0.2336]
! [0.23354, 0.2336]

Final result: eaii!

5. Using the Huffman code in the following figure, decode the encoded string 01010000101011110100. (15 points)

(chapter 8)

a_5 a_6 a_6 a_2 a_5 a_2 a_2 a_2 a_4

Original source			Source reduction			
Symbol	Probability	Code	1	2	3	4
a_2	0.4	1	0.4 1	0.4 1	0.4 1	0.6 0
a_6	0.3	00	0.3 00	0.3 00	0.3 00	0.4 1
a_1	0.1	011	0.1 011	0.2 010	0.3 01	
a_4	0.1	0100	0.1 0100	0.1 011		
a_3	0.06	01010	0.1 0101			
a_5	0.04	01011				

decoding result: $a_5a_6a_6a_2a_5a_2a_2a_2a_4$

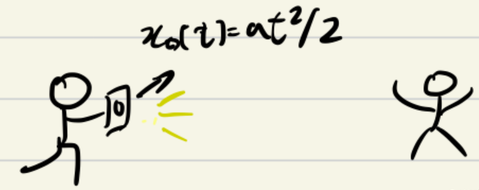
6. Consider the problem of image blurring caused by uniform acceleration in the *x*-direction. If the image is at rest at time t = 0 and accelerates with a uniform acceleration $x_0(t) = at^2/2$ for a time *T*, find the blurring function H(u, v). You may assume that shutter opening and closing times are negligible. (15 points)

CH.5

In motion Blur model

$$H(u,v) = \frac{T}{\pi(ua+vb)} \cdot \sin[\pi(ua+vb)]\, e^{-j\pi(ua+vb)}$$

$x_0(t) = at^2/2$



suppose $\pi ua = n$, apply Fresnel integral

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}$$

$T = 2uaf^2$??

$$C(T) = \int_0^T \cos\left(\frac{\pi}{2} \cdot \frac{2ua\, f^2}{2\pi ua f^2}\right)$$

$C(x) =$

Fresnel $C(z) = \int_0^z \cos(\frac{\pi}{2} \cdot t^2)\, dt$, Fresnel $S(z) = \int_0^z \sin(\frac{\pi}{2} \cdot t^2)\, dt$

$x(z) = \int_0^z \cos(t^2)\, dt$, $y(z) = \int_0^z \sin(t^2)\, dt$

Ans) $H(u,v) = \int_0^T e^{-j2\pi u x_0(t)}\, dt$

$$= \int_0^T e^{-j2\pi u \cdot at^2/2}\, dt$$

$$= \int_0^T e^{-j\pi ua t^2}\, dt \qquad e^{-ix} = \cos x - i \sin x$$

$$= \int_0^T [\cos(\pi ua t^2) - j\sin(\pi ua t^2)]\, dt$$

Apply Fresnel equation

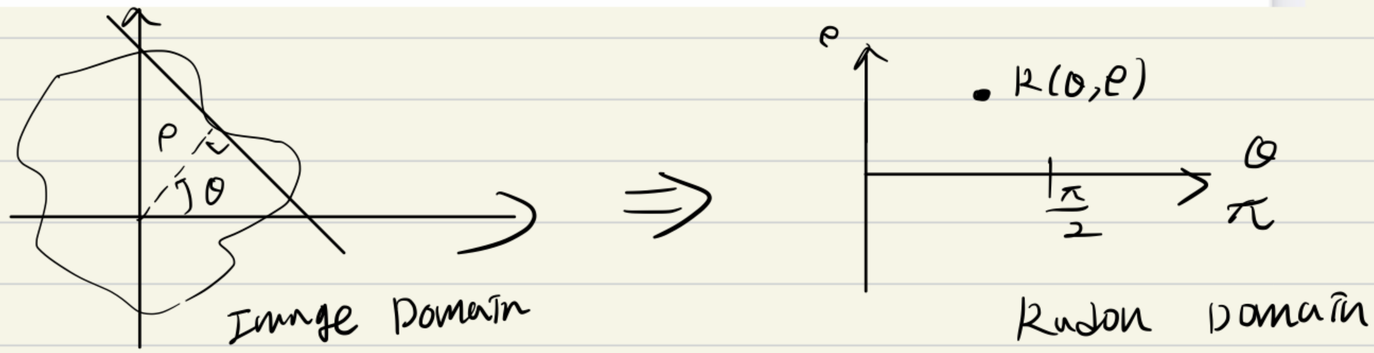
$$= \frac{1}{\sqrt{2\pi ua}} \left(\int_0^{\sqrt{\pi ua}\, T} \cos t^2 dt - j \int_0^{\sqrt{\pi ua}\, T} \sin t^2 dt \right)$$

$$= \sqrt{\frac{1}{2\pi ua}} \left[C(\sqrt{\pi ua}\, T) - j S(\sqrt{\pi ua}\, T) \right]$$

$$C(z) = \sqrt{\frac{2\pi}{T}} \int_0^z \cos t^2 dt$$

$$S(z) = \sqrt{\frac{2}{\pi}} \int_0^z \sin t^2 dt$$

7. Prove that the Radon transform is a linear operator. (10 points)



consider Radon transfer with linear operator

$R(af_1+bf_2) = aR(f_1) + bR(f_2)$ ① a, b is scailing factors

Fourier transform of R(f) in f(x,y)

$* \; G(\omega,\theta) = \int_{-\infty}^{\infty} g(\rho,\theta) e^{-j2\pi\omega\rho} d\rho$

$g(\rho,\theta) = \iint f(x,y)\,\delta(x\cos\theta + y\sin\theta - \rho)\,dxdy$

$R(f) = \iint f(x,y)\,\delta(x\cos\theta + y\sin\theta - \rho)\,dxdy$ ②

For f_1 and f_2

$R(f_1) = \iint f_1(x,y)\,\delta(x\cos\theta + y\sin\theta - \rho)\,dxdy$ ③

$R(f_2) = \iint f_2(x,y)\,\delta(x\cos\theta + y\sin\theta - \rho)\,dxdy$ ④

Consider $f = af_1 + bf_2$

$R(af_1+bf_2) = \iint [af_1(x,y) + bf_2(x,y)]\,\delta(x\cos\theta + y\sin\theta - \rho)\,dxdy$

$= a\iint f_1(x,y)\,\delta(x\cos\theta + y\sin\theta - \rho)\,dxdy + b\iint f_2(x,y)\,\delta(x\cos\theta + y\sin\theta - \rho)\,dxdy$

From ①, ③, ④

$R(af_1+bf_2) = aR(f_1) + bR(f_2)$ Thus, Radon Transform is linear operation